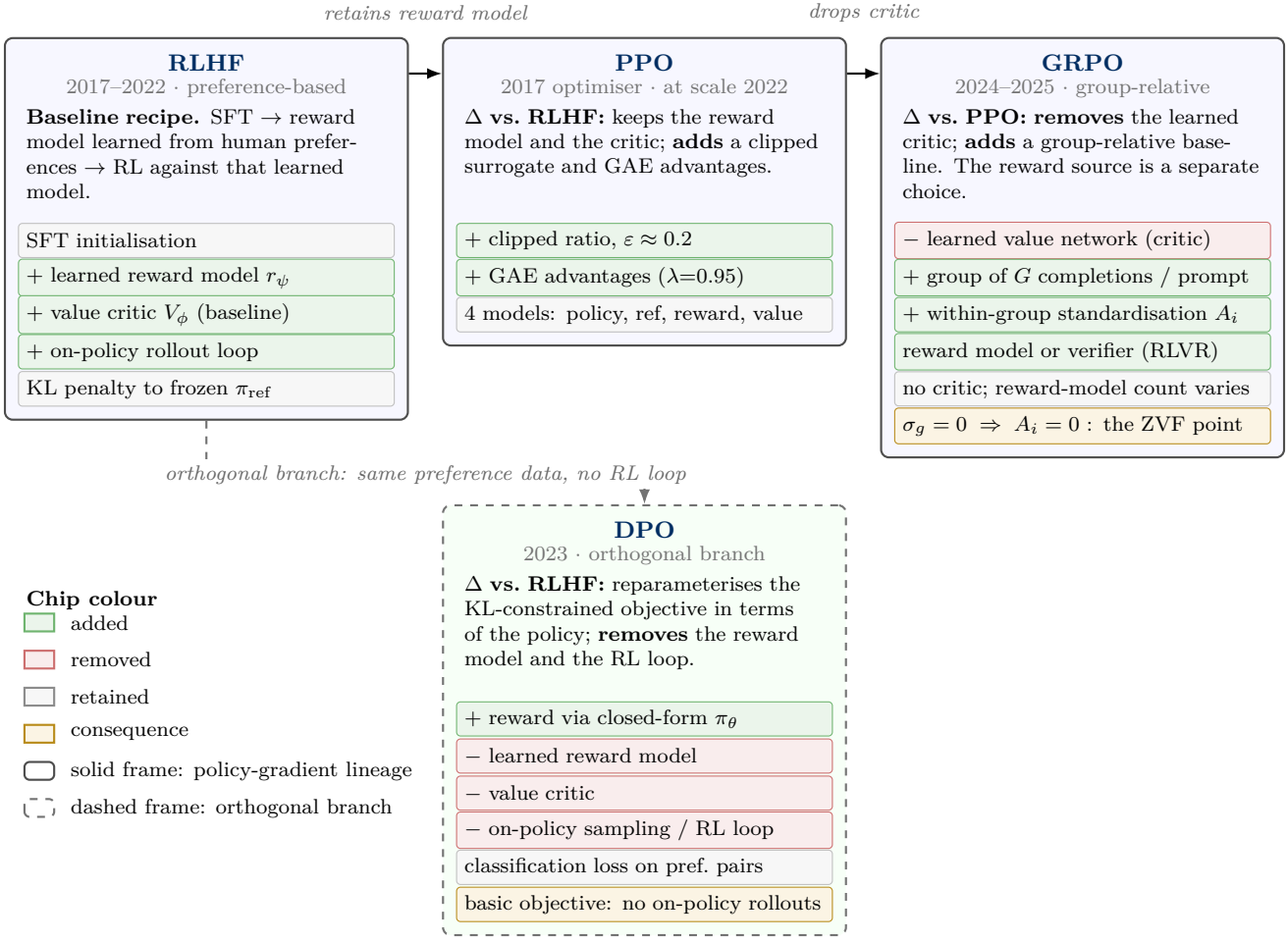
retains reward model
drops critic
RLHF
2017–2022 · preference-based
Baseline recipe. SFT → reward model learned from human preferences → RL against that learned model.
SFT initialisation
+ learned reward model r_ψ
+ value critic V_ϕ (baseline)
+ on-policy rollout loop
KL penalty to frozen π_{ref}
PPO
2017 optimiser · at scale 2022
Δ vs. RLHF: keeps the reward model and the critic; adds a clipped surrogate and GAE advantages.
+ clipped ratio, $\varepsilon \approx 0.2$
+ GAE advantages (λ=0.95)
4 models: policy, ref, reward, value
GRPO
2024–2025 · group-relative
Δ vs. PPO: removes the learned critic; adds a group-relative baseline. The reward source is a separate choice.
− learned value network (critic)
+ group of G completions / prompt
+ within-group standardisation A_i
reward model or verifier (RLVR)
no critic; reward-model count varies
$\sigma_g = 0 \Rightarrow A_i = 0$: the ZVF point
orthogonal branch: same preference data, no RL loop
DPO
2023 · orthogonal branch
Δ vs. RLHF: reparameterises the KL-constrained objective in terms of the policy; removes the reward model and the RL loop.
+ reward via closed-form π_θ
− learned reward model
− value critic
− on-policy sampling / RL loop
classification loss on pref. pairs
basic objective: no on-policy rollouts
Chip colour
added
removed
retained
consequence
solid frame: policy-gradient lineage
dashed frame: orthogonal branch